

Online C Compiler

Practice C problems from our roadmap

C

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     char text[100];
6     int key, i;
7
8     printf("Enter the text: ");
9     fgets(text, sizeof(text), stdin);
10
11     printf("Enter the key (1-25): ");
12     scanf("%d", &key);
13
14     for(i = 0; text[i] != '\0'; i++) {
15         if(text[i] >= 'A' && text[i] <= 'Z') {
16             text[i] = ((text[i] - 'A' + key) % 26) + 'A';
17         }
18         else if(text[i] >= 'a' && text[i] <= 'z') {
19             text[i] = ((text[i] - 'a' + key) % 26) + 'a';
20         }
21     }
22
23     printf("Encrypted Text: %s", text);
24
25     return 0;
26 }
27
28
```

Run

hello
3

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.744 Mb

Sample Input

hello
3

Your Output

Enter the text: Enter the key (1-25): Encrypted Text: khoox

33°

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Online C Compiler

Practice C problems from our roadmap

C

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     char text[100];
6     int a, b, i;
7
8     printf("Enter plaintext (uppercase): ");
9     scanf("%s", text);
10
11    printf("Enter value of a: ");
12    scanf("%d", &a);
13
14    printf("Enter value of b: ");
15    scanf("%d", &b);
16
17    for(i = 0; text[i] != '\0'; i++) {
18        if(text[i] >= 'A' && text[i] <= 'Z') {
19            text[i] = ((a * (text[i] - 'A') + b) % 26) + 'A';
20        }
21    }
22
23    printf("Ciphertext: %s\n", text);
24
25    return 0;
26 }
```

Run

HELLO
5
8

Output

Status : Successfully executed

Time: 0.0000 secs | Memory: 1.872 Mb

Sample Input

HELLO
5
8

Your Output

Enter plaintext (uppercase): Enter value of a: Enter value of b: Ciphertext: RCLLA

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C

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 int main() {
6     char plaintext[100], key[100];
7     int i, j = 0;
8
9     printf("Enter the plaintext: ");
10    scanf("%s", plaintext);
11
12    printf("Enter the key: ");
13    scanf("%s", key);
14
15    int keylen = strlen(key);
16    printf("Ciphertext: ");
17
18    for(i = 0; plaintext[i] != '\0'; i++) {
19        char ch = toupper(plaintext[i]);
20        char k = toupper(key[j % keylen]);
21
22        if(ch >= 'A' && ch <= 'Z') {
23            ch = ((ch - 'A') + (k - 'A')) % 26 + 'A';
24        }
25
26        printf("%c", ch);
27        j++;
28    }
29
30    return 0;
31 }
```

Run

HELLO

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 1.744 Mb

Sample Input

HELLO

Your Output

Enter the plaintext: Enter the key: Ciphertext: GOKKN

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```
16
17 for (int i = 0; i < 5; i++) {
18     for (int j = 0; j < 5; j++) {
19         if (matrix[i][j] == ch) {
20             *row = i;
21             *col = j;
22             return;
23         }
24     }
25 }
26
27
28 int main() {
29     char text[100];
30     int r1, c1, r2, c2;
31
32     printf("Enter plaintext (2 letters): ");
33     scanf("%s", text);
34
35     text[0] = toupper(text[0]);
36     text[1] = toupper(text[1]);
37
38     findPosition(text[0], &r1, &c1);
39     findPosition(text[1], &r2, &c2);
40
41     if (r1 == r2) {
42         printf("Ciphertext: %c%c\n",
43             matrix[r1][(c1 + 1) % 5],
44             matrix[r2][(c2 + 1) % 5]);
45     }
46     else if (c1 == c2) {
47         printf("Ciphertext: %c%c\n",
48             matrix[(r1 + 1) % 5][c1],
49             matrix[(r2 + 1) % 5][c2]);
50     }
51     else {
52         printf("Ciphertext: %c%c\n",
53             matrix[r1][c2],
54             matrix[r2][c1]);
55     }
56     return 0;
57 }
```

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 1.816 Mb

Sample Input

HI

Your Output

Enter plaintext (2 letters): Ciphertext: BF

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hp

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Online C Compiler

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 char matrix[5][5] = {
6     {'M', 'O', 'N', 'A', 'R'},
7     {'C', 'H', 'Y', 'B', 'D'},
8     {'E', 'F', 'G', 'I', 'K'},
9     {'L', 'P', 'Q', 'S', 'T'},
10    {'U', 'V', 'W', 'X', 'Z'}
11 };
12
13 void findPosition(char ch, int *row, int *col) {
14     if (ch == 'J')
15         ch = 'I';
16
17     for (int i = 0; i < 5; i++) {
18         for (int j = 0; j < 5; j++) {
19             if (matrix[i][j] == ch) {
20                 *row = i;
21                 *col = j;
22                 return;
23             }
24         }
25     }
26 }
27
28 int main() {
29     char text[100];
30     int r1, c1, r2, c2;
31
32     printf("Enter plaintext (2 letters): ");
33     scanf("%s", text);
34
35     text[0] = toupper(text[0]);
36     text[1] = toupper(text[1]);
37 }
```

Run

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.816 Mb

Sample Input

HI

Your Output

Enter plaintext (2 letters): Ciphertext: BF

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Online C Compiler

Practice C problems from our roadmap

C



Run

hello

Output

Status : Successfully executed

Time: 0.0000 secsMemory: 1.716 Mb

Sample Input

hello

Your Output

Enter the plaintext: Ciphertext: itssg

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     char plaintext[100], ciphertext[100];
6     char key[] = "QWERTYUIOPASDFGHJKLZXCVBNM";
7     int i;
8
9     printf("Enter the plaintext: ");
10    scanf("%s", plaintext);
11
12    for(i = 0; plaintext[i] != '\0'; i++) {
13        if(plaintext[i] >= 'A' && plaintext[i] <= 'Z')
14            ciphertext[i] = key[plaintext[i] - 'A'];
15        else if(plaintext[i] >= 'a' && plaintext[i] <= 'z')
16            ciphertext[i] = key[plaintext[i] - 'a'] + 32;
17        else
18            ciphertext[i] = plaintext[i];
19    }
20
21    ciphertext[i] = '\0';
22    printf("Ciphertext: %s\n", ciphertext);
23
24    return 0;
25 }
26
```

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